Chapter 11: Architecture Domains

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 11 (pp. 279-294)

Overview

Chapter 11 (pp. 279-294) explains the four architecture domains that every architect works in and connects them to the roles from Chapter 10. Section 11.1 defines Business, Data, Application, and Technology Architecture, notes the history of an "Information" domain, introduces the Interoperability Model, and states that a *complete* EA must address all four domains (pp. 280-286). Section 11.2 separates **domains** (areas of concern) from **layers** (levels of abstraction inside a system), and introduces cross-cutting concerns such as security, motivation, and governance (pp. 286-289). Section 11.3 maps roles to domains (pp. 289-292). Section 11.4 summarizes (pp. 292-294). Leave with this idea: if you skip a domain, what you produce is by definition not a complete Enterprise Architecture, and management must understand the risks of that choice (pp. 285-286). Chapter 11 is part of the PINK and GREEN reading tours (p. 279).

> **Reading note:** Domain definitions and the "complete EA" rule are from the Standard. Jager's views: Information Architecture may return (pp. 282, 285); data + context = information (p. 284); his critique that ACMM treats security only as IT security (p. 288); and his role-to-domain scope figures (pp. 290-292). Small text slips: p. 285 says the "2008 eighth edition", but Ch. 2 dates TOGAF 8 to 2002 (p. 22), and the Summary calls the domains "roles" (p. 292).

Detailed Section Breakdown

Chapter Opener (p. 279)

- **Roles and capabilities recap.** Chief/Enterprise Architect supports strategy, Segment Architect supports portfolios (and sometimes large projects), Solution Architect supports solutions. Every architect deals with one or more domains (p. 279).

11.1 Four Architecture Domains (pp. 280-286)

Domain	Definition	Page
Business Architecture	Business strategy, governance, organization, and key business capabilities, so IT supports business objectives	p. 280
Data Architecture	Structure of logical and physical data assets and data management resources	p. 280
Application Architecture	Blueprint of individual systems, their interactions, and their relationships to core business processes	p. 281
Technology Architecture	Hardware, software, and network infrastructure supporting mission-critical applications	p. 281

- **Information Systems Architectures.** The collective name for Data and Application Architecture (p. 281, Figure 11-1).
- **History.** Four domains since TOGAF 8. TOGAF 7 recognized only Technology Architecture. TOGAF 8 called the data domain "Data *or Information* Architecture", and the "or Information" was later dropped (pp. 281-282).

- **Interoperability Model (Figure 11-2).** A five-layer model used in healthcare. It splits Business into organization and processes and uses Information instead of Data. Otherwise it matches TOGAF's four domains, with legislation/regulation and security as preconditional columns (pp. 282-283).
- **Data entity vs. information concept.** TOGAF's Data Architecture has the data entity but no information concept, while BIZBOK describes information concepts in much more detail. The two are not comparable (pp. 283-284).
- **Data + Context = Information.** "1 Primary Lane" means little until you add that it is the President's home address. A list of numbers becomes information once you know they are next week's temperatures (p. 284).
- **Complete EA rule.** "A complete Enterprise Architecture should address all four architecture domains." Time and resource limits often prevent this, and architecture descriptions are scoped to purpose in the first ADM phase (p. 285).
- **Partial architectures.** If business processes won't change, a full Business Architecture may not be needed, but it must still be understood, because Data, Application, and Technology build on it (p. 285).
- **Risk of incompleteness.** Inconsistency and integration problems, with later costs. The architect must spell out the risks and trade-offs so management understands them (pp. 285-286).

Jager's commentary (author's view): With growing BIZBOK integration, the Information Architecture domain may reappear in TOGAF (pp. 282, 285).

11.2 Difference in Domains and Layers (pp. 286-289)

- **Organizations' part.** Architects translate strategy into execution, but organizations must define their strategy and work with architects on feasible plans. EA helps shape drivers, goals, objectives, and initiatives (p. 286).
- **Broad scope.** Organizational design, process design, information flow visualization, and so on. EA, once technology-focused (around 1990), now covers all aspects of an organization (p. 286).

	Architecture domains	Architecture layers
What	Broad divisions: areas of concern or expertise in the overall architecture	Levels of abstraction within a system or Solution Architecture
Examples	Business, Data, Application, Technology	Presentation, business logic, data access, infrastructure (Figure 11-3)
Purpose	Categorize the concerns and perspectives to address	Organize components and functionality, communicating via interfaces, for modularity and reuse

(pp. 287-289)

- **Complementary.** Layers organize systems *within* a domain (p. 289).
- **Cross-cutting concerns (Figure 11-4).** Security (sometimes called a domain, but it cuts across all of them), motivation, and governance (pp. 288-289).

Jager's commentary (author's view): The ACMM maturity model (Ch. 15) lists IT security as one of nine elements, which ignores that security is cross-cutting (p. 288).

11.3 Mapping Roles to Domains (pp. 289-292)

Role	Domain scope	Figure / page
Chief/Enterprise Architect	The entire ecosystem: all four domains. A Chief Architect focuses on strategy. Without one, the EA does both	Figure 11-6, p. 290

Role	Domain scope	Figure / page
Segment Architect	One domain per specialization: Business, Data, Application, or Technology Architect	Figure 11-7, p. 291
Solution Architect	A defined (usually technical) project. "Nine out of ten" times within Technology. Specializations: Cloud, Network, Software	Figure 11-8, pp. 291-292

- Practice differs, but recognize the difference between roles and specializations and don't hold false expectations of a role (p. 292).

11.4 Summary (pp. 292-294)

- The four domains, the Interoperability Model alternative, and the complete-EA rule (pp. 292-293).
- Domains vs. layers. Security, motivation, and governance as cross-cutting concerns (p. 293).
- Role-domain mapping: EA covers all four, Segment Architects one each, Solution Architects mostly Technology (p. 294).

Key Terms & Definitions

Term	Definition	Page
Business Architecture	Business strategy, governance, organization, key capabilities	p. 280
Data Architecture	Logical and physical data assets and data management resources	p. 280
Application Architecture	Blueprint of systems, interactions, and relation to core processes	p. 281
Technology Architecture	Hardware, software, and network infrastructure	p. 281
Information Systems Architectures	Collective name for Data and Application Architecture	p. 281
Interoperability Model	Five-layer model (healthcare) splitting business into organization and process, with an information layer	pp. 282-283
Architecture layer	A logical division of a software system by function (presentation, logic, data access, infrastructure)	p. 287
Cross-cutting concern	A concern spanning all domains: security, motivation, governance	pp. 288-289

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 11-1	The four core architecture domains	p. 280
Figure 11-2	Interoperability Model (five-layer model)	p. 283
Figure 11-3	Architecture layers	p. 287
Figure 11-4	Domains and cross-cutting concerns	p. 289
Figure 11-5	Roles and specializations vs. domains	p. 290
Figures 11-6 to 11-8	Scope of the Chief/Enterprise, Segment, and Solution Architect	pp. 290-292

Self-Check Questions

Q: Which two domains make up the Information Systems Architectures?

A: Data Architecture and Application Architecture (p. 281).

Q: Can an architecture that covers only three domains be called a complete Enterprise Architecture?

A: No. By definition a complete EA addresses all four domains. Partial work risks inconsistency and later integration costs, which must be explained to management (pp. 285-286).

Q: If business processes won't change, can you skip Business Architecture?

A: A comprehensive Business Architecture may not be needed, but it must still be considered and understood, because the other domains build on it (p. 285).

Q: How do architecture domains differ from architecture layers?

A: Domains are broad areas of concern across the whole architecture. Layers are levels of abstraction inside a system (presentation, logic, data access, infrastructure) (pp. 287-289).

Q: Name three cross-cutting concerns.

A: Security, motivation, and governance (pp. 288-289).

Q: Explain "Data + Context = Information" with the book's example.

A: "1 Primary Lane" is meaningless raw data until you add the context that it is the President's home address. Then it becomes information (p. 284).

Q: How does the Interoperability Model differ from TOGAF's four domains?

A: It splits Business into organization and processes and uses Information instead of Data. Otherwise it matches TOGAF's four domains (pp. 282-283).

Q: Which domain does a Solution Architect usually work in, and why?

A: Technology, because the scope is usually a defined technical project (p. 291).

Q: What did TOGAF 8 call the data domain, and why does Jager find that interesting?

A: "Data or Information Architecture". The "or Information" was later dropped, but BIZBOK integration may bring Information Architecture back (pp. 282, 285).